

Related Rates



Setting up related-rates problems

- 1 A circular oil spill is expanding so that its radius increases at 3 m/min. How fast is the area increasing when the radius is 50 m?
- 2 A 10 m ladder leans against a wall. The bottom slides away from the wall at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?

Volume and other rates

- 3 Air is pumped into a spherical balloon at $100 \text{ cm}^3/\text{s}$. Find the rate of change of the radius when $r = 5 \text{ cm}$, to three decimal places.
- 4 Water drains from a conical tank (vertex down) with radius 4 m and height 10 m at the top, at a rate of $2 \text{ m}^3/\text{min}$. Using similar triangles $r = \frac{2h}{5}$, find how fast the water level is dropping when $h = 5 \text{ m}$.